SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT ENGINEERING

Program: B.Tech /MBA Tech (Data Science) Year: II Semester : III

AcademicYear: 2022-2023

Subject: Managing Uncertainty

Date : 30 January 2023

Marks: 100

Time: 10.00 am - 1.00 pm

Durations: 3 (Hrs)

No. of Pages: 05+1 = 6

Re-Examination (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

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Q1		Answer briefly:	20
CO-1 ; SO-2 ; BL-1	A	The number of calls coming per minute into a hotels reservation is Poisson random variable with mean 3 Find the probability that no calls come in given 1 minute period	5
CO-2 ; SO-2 ; BL-2	B	Explain Naive Bayes theorem	5

<table>
<tr><td>CO-3 ;
SO-1 ;

BL-2</td><td>C</td><td>I. mean = 100 , SD = 20, mode = 80. Find Karl Pearson coefficient of Skewness
II. mean = 50, variance = 400, Karl Pearson coefficient of Skewness = -0.4.
Find mode</td><td>3
2</td></tr>
<tr><td>CO- 3;
SO-2 ;

BL-1</td><td>D</td><td>These data are a sample of the number of seriously ill patients treated for the past five weeks in a hospital:
33 50 22 27 48
Calculate the variance and standard deviation for the given data.</td><td>5</td></tr>
<tr><td>Q2</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q2

CO-3;
SO-1;

BL-1</td><td>A</td><td>Keyes Home Furnishings ran six local newspaper advertisements during December. The following frequency distribution resulted;

No. of times subscribers saw AD during December 0 1 2 3 4 5 6
Frequency 897 1,082 1,325 814 307 253 198

What is the average number of times a subscriber saw a Keyes ad during December?</td><td>10</td></tr>
<tr><td>Q2

CO-3;
SO-1;

BL-2</td><td>B</td><td>The Fish and Game station on Lake Wylie keeps records of fish caught on the lake and reports its finding to the National Fish and Game Service. The catch in pounds for the last 20 days was:

<table><tr><td>101</td><td>132</td><td>145</td><td>144</td><td>130</td><td>88</td><td>156</td><td>188</td><td>169</td><td>130</td></tr><tr><td>90</td><td>140</td><td>130</td><td>139</td><td>99</td><td>100</td><td>208</td><td>192</td><td>165</td><td>216</td></tr></table>
Calculate the range, variance, and standard deviation for these data. In this instance, is the range a good measure of the variability? Why?</td><td>10</td></tr>
<tr><td>Q3</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q3</td><td></td><td>Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination.
Scores on the GMAT are roughly normally distributed with a mean of 527</td><td>10</td></tr>
</table>

CO-2; SO-2; BL-2	A	and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT? How high must an individual score on the GMAT in order to score in the highest 5%?	
Q3 CO-2; SO-1; BL-2	B	A multiple choice exam has 4 choices for each question. A student has studied enough so that the probability they will know the answer to a question is 0.5, the probability that they will be able to eliminate one choice is 0.25, otherwise all 4 choices seem equally plausible. If they know the answer they will get the question right. If not they have to guess from the 3 or 4 choices. As the teacher you want the test to measure what the student knows. If the student answers a question correctly what's the probability they knew the answer?	10
Q4		Answer briefly:	20
Q4 CO-;3 SO-2; BL-1	A	Distribution of monthly income of a group of 3000 school teachers in a village conforms to a normal curve with $\mu = 600$ and $\sigma = 100$. I. Find the percentage of teachers having monthly income of more than Rs. 800. II. Number of teachers having monthly income less than Rs. 400. III. Find the lowest monthly income amongst the highest paid 100 teachers. IV. The highest monthly income among the lowest paid teachers. V. Calculate the range of 3σ	10
Q4 CO-;3 SO-2; BL-1	B	Given a binomial distribution with n = 25 and p = 0.032, use the Poisson approximation to the binomial to find I. P(r=3) II.P(r=5) III.P($r \leq 2$)	3 3 4
Q5		Answer briefly:	20

Q5 CO-3; SO-2; BL-2	A	Of the students in the college, 60% of the students reside in the hostel and 40% of the students are day scholars. Previous year results report that 30% of all students who stay in the hostel scored A Grade and 20% of day scholars scored A grade. At the end of the year, one student is chosen at random and found that he/she has an A grade. What is the probability that the student is a hosteler?	10
Q5 CO-3; SO-2; BL-2	B	A brokerage survey reports that 30 percent of individual investors have used a discount broker, that is, one which does not charge the full commission. In a random sample of 9 individuals, what is the probability that I. Exactly two of the sampled individuals have used a discount broker II. Not more than three have used a discount broker III. At least three of them have used a discount broker	4 4 2
Q6		Answer briefly:	20
Q6 CO-3; SO-2; BL-2	A	There are five machines in a factory. Of these machines, three are working properly and two are defective. Machines which are working properly produce articles each of which has independently a probability of 0.1 of being imperfect. For the defective machines this probability is 0.2. A machine is chosen at random and five articles produced by the machine are examined	10
Q6 CO-3; SO-2; BL-2	B	Suppose now that events A, B and C are mutually independent with P(A) = 0.3, P(B) = 0.4, P(C) = 0.5. Compute the following: (Hint: Use a Venn diagram) I. $P(A \cap B \cap C^c)$ II. $P(A \cap B^c \cap C)$ III. $P(A^c \cap B \cap C)$	3 3 4

<table>
<tr><td>Q7</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q7

CO-2;
SO-2;

BL-2</td><td>A</td><td>Dave's Giveaway Store advertise, "if our average prices are not equal or lower than everyone else's, you get it free". One of Dave's customers came into the store one day and threw on the counter bills of sale for six items she bought from a competitor for an average price less than Dave's. the items cost

$1.29 $2.97 $3.49 $5 $7.50 $10.95

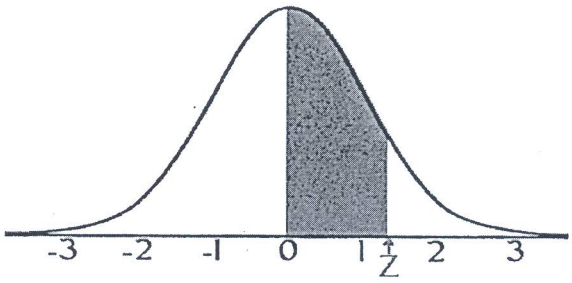
Dave's prices for the same six items are $1.35, $2.89, $3.19, $4.98, $7.59 and $11.50. Dave told customer, "My ad refers to a weighted average price of these items. Our average is lower because our sales of these items have been"

7 9 12 8 6 3</td><td>10</td></tr>
<tr><td>Q7

CO-3;
SO-2;

BL-2</td><td>B</td><td>The frequency distribution below represents the weights in pounds of a sample of packages carried last month by a small airfreight company.

<table><tr><td>10-10.9</td><td>11.0-11.9</td><td>12.0-12.9</td><td>13.0-13.9</td><td>14.0-14.9</td><td>15.0-15.9</td><td>16.0-16.9</td><td>17.0-17.9</td><td>18.0-18.9</td><td>19.0-19.9</td></tr><tr><td>1</td><td>4</td><td>6</td><td>8</td><td>12</td><td>11</td><td>8</td><td>7</td><td>6</td><td>2</td></tr></table>
I. Compute the sample mean
II Compute the sample mean using the coding method with 0 assigned to the fourth class
III Repeat part (II) with 0 assigned to the sixth class.</td><td>10</td></tr>
</table>



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998